

A. PROTOCOL SPECIFICATION

A.1. Sequence grammar

```

Episode ::= Prefix (ToolBlock* Unit)+
Unit ::= UNIT.START Environment ASRView
Decision UNIT.END
ASRView ::= ASR ASRPayload ASR.EOS |  $\epsilon$ 
Decision ::= Listen | ContinueSpeak |
FinalSpeak
Listen ::= LISTEN NextWindow CHUNK.EOS
ContinueSpeak ::= SPEAK Action Text
NextWindow CHUNK.EOS
FinalSpeak ::= SPEAK Action Text NextWindow
TURN.EOS CHUNK.EOS
Action ::= MEMORY | LAUNCH: $N$  | CANCEL: $N$ 
| CANCEL: $N$  LAUNCH: $N$  |  $\epsilon$ ,
 $N \in \{0, 1, 2, 3\}$ 
NextWindow ::= WINDOW.SHORT | WINDOW.MID |
WINDOW.LONG
Text ::= TextToken*
ToolBlock ::= IM.START tool ToolPayload IM.END

```

NextWindow schedules the following Unit rather than changing the current one, so both LISTEN and SPEAK end with a window token and `<|chunk.eos|>`. A speaking turn is a maximal sequence of SPEAK Units terminated by FinalSpeak; adjacent speaking turns need not be separated by LISTEN. SPEAK and text presence are orthogonal, so Text may be empty. Tool Blocks occur only at Unit boundaries, and trailing Tool Blocks without a following Unit are invalid.

Because every behavioral decision is an explicit token, additive logits bias changes runtime behavior without modifying weights. Table 5 lists the three uses.

Table 5. Logits-bias controls. Each adds a constant to the named token’s logit at inference.

Bias target	Effect
<code>< listen ></code> vs. <code>< speak ></code>	Shift toward listening or speaking
An action token, set to $-\infty$	Disable that action
<code>< window:short ></code> ,	Select the next-window duration
<code>< window:mid ></code> ,	
<code>< window:long ></code>	

A.2. Token roles and supervision

A.3. State lifetimes

Thinker history persists across LISTEN/SPEAK transitions and speaking turns. Talker state persists across Units only within one speaking turn and is reset at its terminal SPEAK; a turn reset never clears Thinker episode history.

A.4. Turn bootstrap and text FIFO

Each speaking turn owns one Talker row and one FIFO. Before its first acoustic position, the Talker inserts the five target-free bootstrap conditions in Table 8; continuation Units do not repeat them.

When a SPEAK Unit arrives, its assistant-text tokens are appended in order. A final Unit also appends `<|tts.eos|>`. Every attended acoustic position pops one text condition from the left; an

Table 6. Canonical Thinker ranges and training roles. “Masked” means that the range remains in causal context but has no Thinker target.

Range	Role	Target
Prefix, Tool Block	Episode/external context	Masked
Environment, Unit delimiters	Causal input/boundary	Masked
ASR Block	Optional ASR training view	If enabled
LISTEN/SPEAK	Interaction control	Valid
Action tokens	MEMORY and indexed L2 actions	Valid
Assistant text	Reply content	Valid
Window token	Schedule following Unit	Valid
TURN/CHUNK EOS	Turn/Unit speech boundary	Valid

Table 7. State ownership across the three protocol lifetimes.

Lifetime	Retained state	Reset boundary
Episode	Thinker prefix, causal history, committed Units	Episode close
Unit	Current environment and canonical decision	Commit to history
Speaking turn	Talker KV, text FIFO, codec/PCM rolling state	Final SPEAK

Table 8. Talker bootstrap at the start of each speaking turn.

Position	Text channel	Codec channel
1	<code>< tts.pad ></code>	codec-nothink
2	<code>< tts.pad ></code>	codec-think-BOS
3	<code>< tts.pad ></code>	codec-think-EOS
4	<code>< tts.pad ></code>	speaker identity
5	<code>< tts.bos ></code>	codec-pad

empty queue supplies `<|tts.pad|>`. Unconsumed text therefore carries across Units within the same speaking turn, but neither the FIFO nor Talker KV state crosses a turn boundary. ASR, control, action, window, boundary, and Tool-Block tokens are never enqueued. If the acoustic slots of a turn cannot consume the remaining payload and the final `<|tts.eos|>`, the turn is rejected rather than built with truncated text.

After the bootstrap, the first Unit combines the first FIFO condition with codec BOS and predicts its first real frame. A continuation Unit has no synthetic restart: its predecessor’s last real codec frame stays pending, and after the new Unit’s text is enqueued that frame combines with the next FIFO condition to predict the new Unit’s first frame. The pending frame remains a supervised real frame. No codec BOS, codec PAD, or synthetic boundary is inserted between ordinary Units.

Ending the turn takes two further steps, and neither produces speech. Thinker `<|turn.eos|>` only marks the decision; it enqueues `<|tts.eos|>` so the Talker learns that no text remains. Once the remaining speech has been consumed, the last real codec frame predicts `<|codec.eos|>`; this is the same stopping condition in training and runtime, with no separate `<|codec.pad|>`-conditioned EOS position. The EOS prediction occupies no acoustic frame and carries no residual-code target, so it supervises where speech stops rather than what the next frame sounds like. At runtime `<|codec.eos|>`

is masked in every non-final Unit, so speech cannot stop mid-turn. A final Unit usually holds fewer real frames than its window budget, since a speaking turn rarely ends on a window boundary, so it may predict $\langle \text{codec_eos} \rangle$ before that budget runs out. Only if the budget runs out first does one extra non-acoustic step check the EOS prediction under the same last-real-frame condition; anything other than $\langle \text{codec_eos} \rangle$ is logged and dropped. The chain is therefore $\langle \text{turn_eos} \rangle \rightarrow \langle \text{tts_eos} \rangle \rightarrow \langle \text{codec_eos} \rangle$.

A.5. Text-ahead transformation

For each speaking turn, let T_k be the ordered assistant-text tokens assigned to its k -th Unit. At an internal boundary k , Text-ahead samples a lead $d_k \in [0, d_{\max}]$, with $d_{\max} = 2$ in the reported experiments, and moves the first $\min(d_k, |T_{k+1}|)$ tokens of T_{k+1} to the tail of T_k . Transfers use the original per-Unit lists and occur only between adjacent Units, so applying all boundaries preserves the flattened turn sequence exactly:

$$T_1 \parallel \dots \parallel T_K = T'_1 \parallel \dots \parallel T'_K. \quad (1)$$

The transformation changes only when a token enters the Talker FIFO. It does not change token order or count, LISTEN/SPEAK and action decisions, codec targets, or the causal boundary of future environments. Tokens never cross a speaking-turn boundary. A turn with inconsistent Unit-level supervision masks is left unchanged, because moving a token would change its training ownership.

The random-ahead schedule draws (d_k) from a bounded random walk whose transitions increase, hold, decrease, or reset the current lead, with weights (3, 2, 2, 1); the fixed-ahead schedule sets all internal leads to $d_{\max}$. The transformation is applied to an eligible turn with probability $p_{\text{TA}} = 0.8$ and is disabled when $d_{\max} = 0$.

B. ADAPTIVE WINDOW POLICY

B.1. Prediction target and configurations

At the end of window k , the Thinker predicts a three-way token z_k that schedules window $k+1$, using only multimodal context causally available through the boundary of window k . SHORT, MID, and LONG are an ordered temporal policy, not durations intrinsic to the method. The experimental configuration instantiates them as in Table 9; every duration holds an integer number of 12.5-Hz codec frames. MID initializes the first window and is the fallback when no earlier rule applies.

Table 9. Window configurations used in the experiments.

Token	Duration (s)	Codec frames
SHORT	0.48	6
MID	0.64	8
LONG	0.96	12

B.2. Weak-label priority rules

Weak targets are assigned by a deterministic script over dataset metadata: timestamped acoustic voice-activity boundaries, a complete/incomplete semantic label on each speaking-to-silence boundary, an interruption/backchannel label on each overlap, and the ASR transcript with word timestamps. These annotations are used only to build the target. Training exposes the resulting window token, and inference predicts it from the Thinker state without the annotations.

The script returns at the first matching rule, so priority is the rule order and no separate tie-breaking step is needed. The speech-rate test compares the transcript-derived rate r against a language-specific threshold: $\tau_{\text{zh}}=3.5$ Chinese characters/s and $\tau_{\text{en}}=2.0$ English words/s, each about $0.7\times$ its conversational rate, so ordinary speech selects MID and only distinctly slow speech selects LONG.

Algorithm 1 Next-window weak target at a window boundary.

```

1: IF the window carries an interruption annotation or the trans-
   script contains a correction marker (不对, 等等, wait, actually)
   THEN
2:   RETURN SHORT
3: IF a speaking-to-silence boundary falls in the window THEN
4:   RETURN SHORT IF its semantic label is complete, ELSE
     MID
5: IF the transcript contains a hesitation marker (嗯, 那个, um)
   THEN
6:   RETURN SHORT
7: IF silence is sustained THEN
8:   RETURN MID IF accumulated silence < 1.0 s, ELSE LONG
9: IF a silence-to-speaking transition falls in the window THEN
10:  RETURN MID
11: IF speech is sustained THEN
12:  RETURN MID IF  $r > \tau_{\text{lang}}$ , ELSE LONG
13: IF the window carries a backchannel annotation THEN
14:  RETURN the preceding window's target
15: RETURN MID

```

C. TIERED ACTIONS AND ASYNCHRONOUS REQUESTS

C.1. Action levels

L0, L1, and L2 are action levels, not token names. L0 is the direct response and emits no action token. L1 emits $\langle \text{memory} \rangle$. L2 uses indexed $\langle \text{launch}:N \rangle$ and $\langle \text{cancel}:N \rangle$, $N \in \{0, 1, 2, 3\}$, to control up to four external requests. Action selection is independent of the LISTEN/SPEAK control: action tokens, when present, follow $\langle \text{speech} \rangle$ and precede assistant text, as in the grammar of Appendix A.1. Table 10 summarizes the three paths. None of them blocks the current response.

Table 10. Action levels and their execution paths.

Level	Action token	Execution	Blocking
L0	None	Thinker response	No
L1	$\langle \text{memory} \rangle$	Local backend	No
L2	$\langle \text{launch}:N \rangle, \langle \text{cancel}:N \rangle$	External backend	No

C.2. L1 local state

The minimal L1 backend keeps stable user attributes and preferences in PROFILE and episodic conversational facts in MEMORY. The parameter-free $\langle \text{memory} \rangle$ trigger asks the backend to update that state from recent dialogue; it does not carry field values inside the canonical sequence. The current reply continues without waiting. A successful update is visible to later Units, and a failed update neither blocks the current response nor changes the previously committed state. A richer backend may replace this one provided it keeps the same trigger and timing.

C.3. L2 slot lifecycle

Each of the four slots is independently idle or bound to one in-flight request. Table 11 gives the legal per-slot transitions. NONE is the absence of an L2 action and leaves every slot unchanged. An invalid action is rejected rather than reinterpreted: launch requires an idle slot and cancel requires an occupied one.

Table 11. Per-slot L2 transitions at a Unit decision.

Slot state	Action	Effect	Next state
Idle	NONE	No external action	Idle
Idle	LAUNCH: N	Launch new g	Active(g)
Active(g)	NONE	Keep request	Active(g)
Active(g)	CANCEL: N	Cancel g	Idle
Active(g)	CANCEL: N ; LAUNCH: N	Replace g with g'	Active(g')

Requests run while full-duplex interaction continues. A cancel invalidates the slot’s active identity before cancellation finishes, so a result that races the cancellation cannot re-enter as current. Same-slot replacement cancels the old identity before launching the new one. A result is accepted only when its slot, request identity, and stage index are still current; duplicates and out-of-order results are discarded.

C.4. Return path

A current request may return one result or an ordered sequence of stages. Each stage is a complete semantic unit; partial token fragments are never shown to the Thinker. An accepted stage, whether a valid result or an explicit backend failure, is serialized as a `tool`-role Tool Block, given the next stage index, and inserted at a later Unit boundary. As defined in Appendix A, the block is loss-masked but stays in the causal context, which is how the Thinker distinguishes a reported failure from silence. Cancelled, stale, and unreturned requests insert nothing. Tool Blocks enter the Thinker context only and are not queued in the Talker FIFO. No return path blocks the current response. The protocol specifies no production timeout or retry; an expired or failed backend is reported, when it is reported, through the failure block above.

D. PROGRESSIVE TRAINING DETAILS

The data-difficulty axis (C1/C2/C3) is orthogonal to the parameter axis (Thinker-only, Talker-only, joint): only Thinker SFT traverses the curriculum, while the Talker-only and joint stages each use one fixed data mix.

D.1. Data generation pipeline

Training episodes come from an automated multi-agent pipeline. Text-only dialogue outlines are generated by orchestrator and participant LLM agents, then rendered into synthetic speech with timestamp alignment. Quality control combines ASR round-trip verification, overlap sanity checks, and human spot audits at each curriculum stage. The pipeline is the subject of a companion study; this section states only what the training setup consumes.

D.2. Thinker curriculum SFT

Table 12 lists the three stages. Duplex coverage increases from C1 to C3, and the stage episode sets are disjoint, so a later stage does not reuse an earlier stage’s episodes and no mixing weight is involved.

Each stage runs one epoch at a global batch of 256; a trailing partial batch counts as one update.

Table 12. Thinker curriculum stages.

	C1	C2	C3
Scope	Turn-based	Basic duplex	Full duplex
Window sup.	None (fixed MID)	Dynamic	Dynamic
Actions	None	None	L1 and L2
Duplex events	None	BC, pause, simple int.	All events
Episodes	63,659	190,976	509,269
Updates	249	746	1,990

D.3. Talker training objective

Talker-only SFT freezes the Thinker and trains the Talker together with its residual CodePredictor. Teacher forcing feeds the ground-truth codec sequence, so each position conditions on the preceding real frames. The main head predicts group 0 of the current frame; the CodePredictor predicts residual groups 1–15, giving fifteen residual targets per frame. As specified in Appendix A, the final position of a speaking turn predicts `<|codec.eos|>` from the last real frame and carries no residual target. No curriculum is used: codec generation conditioned on text and Thinker context is the same task at every difficulty.

D.4. Loss weighting and joint SFT

A per-token-class weight scales the loss numerator; the denominator counts supervised tokens with no weighting. Both the weighted sum and the count are reduced across data-parallel workers before the division. Tool Block tokens are always masked, since they enter from an external source and serve only as context. Table 13 gives the weights.

Table 13. Per-token-class loss weights.

Token class	Thinker-only	Joint
Control (<code>< listen ></code> / <code>< speak ></code>)	2.0	2.0
Window token	1.5	1.5
Action token	1.5	1.5
Assistant text	1.0	1.0
Boundary token	4.0	4.0
ASR view (when enabled)	1.0	1.0
Tool Block	masked	masked

The weight scales the loss numerator. The denominator counts supervised tokens without weighting.

Let S be a weighted sum of per-token losses and N the matching unweighted count. Joint SFT continues on a smaller high-quality set with every component trainable:

$$\begin{aligned}
 \mathcal{L}_{\text{thinker}} &= S_{\text{thinker}}/N_{\text{thinker}}, \\
 \mathcal{L}_{\text{talker}} &= S_{\text{main}}/N_{\text{main}} + 0.3 S_{\text{res}}/N_{\text{res}}, \\
 \mathcal{L}_{\text{joint}} &= \mathcal{L}_{\text{talker}} + 3 \mathcal{L}_{\text{thinker}}.
 \end{aligned} \tag{2}$$

D.5. Optimization

Table 14 collects the optimizer setup. The three SFT stages share it; GRPO does not inherit it.

D.6. Policy optimization (GRPO)

GRPO continues from the joint-SFT checkpoint and updates the interaction policy without changing the architecture. For each episode,

Table 14. Optimizer setup for the SFT stages and GRPO.

	SFT	GRPO
Optimizer	Adam ($\beta_1=0.9, \beta_2=0.999$)	Adam ($\beta_1=0.9, \beta_2=0.95$)
Peak learning rate	1×10^{-4}	1×10^{-5}
Global batch size	256	8
Precision	bfloat16	bfloat16
Hardware	8 × NVIDIA H100	8 × NVIDIA H100
LR schedule	cosine	cosine
Warmup ratio	0.05	0
Max sequence length	32,768	32,768
Gradient clipping	1.0	1.0
Weight decay	0.01	0.1

G rollouts are sampled and the advantage of each is its reward relative to the others in the group, so no value function is trained.

The reward has three separately evaluated dimensions.

- **Conversational rhythm.** Penalties for late responses, premature interruptions, and missed backchannels, computed from the streaming timestamps with no human annotation.
- **Response quality.** Speech-stream naturalness from a UTMOSv2 five-fold ensemble, plus text-stream relevance from a reference-free LLM judge.
- **L2 action appropriateness.** An LLM judge scores whether each $\langle \text{launch}: N \rangle$ and $\langle \text{cancel}: N \rangle$ is justified by the preceding context. L0 and L1 are the implicit baselines, so unnecessary and missed triggers are both penalized.

The composite reward is $\alpha_1 r_{\text{rhythm}} + \alpha_2 r_{\text{quality}} + \alpha_3 r_{\text{L2}}$. Table 15 lists the GRPO-specific quantities.

Table 15. GRPO configuration.

Quantity	Value
Rollouts per episode G	8
Reward weights $\alpha_1, \alpha_2, \alpha_3$	equal
KL coefficient	0

E. EXPERIMENTAL DETAILS AND ADDITIONAL RESULTS

E.1. Complete full-duplex results

Table 18 gives the per-scenario results behind the aggregated main-text numbers, for every reported training stage and window policy. Panel (a) is FDB-v1 and panel (b) is FDB-v1.5.

E.2. Serialization and window-policy execution

Table 19 reports the token and decoding costs of the serialization formats and the per-window costs of the window policies, including the SPEAK/LISTEN decomposition.

E.3. Text-ahead EOS distributions

Table 16 breaks the EOS-offset distribution down by language for the three Text-ahead schedules. The offset bins are $[0, 1)$ for ValidEnd, $[1, 3)$ and $[3, 5)$ for the intermediate late detail, $[5, 16)$ for Late5+, and $[16, +\infty)$ for NoEOS.

E.4. L1 and L2 diagnostics

Action prediction and information utilization share the 1,000 evaluation episodes but score different views. Action prediction expands

Table 16. Text-ahead speech quality and EOS-offset distributions.

(a) Language-specific speech quality						
Group	Lang.	N	UTMOSv2 $\uparrow$	Mixed UTMOSv2 $\uparrow$	CER (%) $\downarrow$	WER (%) $\downarrow$
Aligned	ZH	1,516	2.792 $\pm$.079	2.902 $\pm$.145	10.74	–
	EN	1,514	3.012 $\pm$.107		–	19.10
Fixed Ahead	ZH	1,516	3.428 $\pm$.043	3.555 $\pm$.135	5.38	–
	EN	1,514	3.682 $\pm$.051		–	10.03
Random Ahead	ZH	1,516	3.513 $\pm$.077	3.648 $\pm$.160	4.83	–
	EN	1,514	3.783 $\pm$.092		–	8.93

(b) Termination-offset distribution

Group	Lang.	N	$[0, 1)$ (%)	$[1, 3)$ (%)	$[3, 5)$ (%)	$[5, 16)$ (%)	$[16, +\infty)$ (%)	Mean trunc. tokens
Aligned	ZH	1,516	51.25	37.86	8.71	2.18	0.00	0.9321
	EN	1,514	48.75	22.59	14.07	13.87	0.73	1.9478
Fixed Ahead	ZH	1,516	87.27	10.82	1.52	0.40	0.00	0.2078
	EN	1,514	82.23	9.58	4.76	3.30	0.13	0.5436
Random Ahead	ZH	1,516	90.04	8.71	1.12	0.13	0.00	0.1563
	EN	1,514	82.69	9.78	4.62	2.84	0.07	0.4808

Panel (a): N counts evaluated utterances. UTMOSv2 is the mean $\pm$ standard deviation; CER applies to ZH and WER to EN, so the other cell is –, and Mixed UTMOSv2 pools both. Panel (b): entries are percentages of the language-specific turns, and mean trunc. tokens uses the same subset.

Table 17. L1 and L2 action prediction.

(a) L1 action prediction							
Scope	Decision points	TP	FP	FN	Prec. $\uparrow$	Rec. $\uparrow$	F1 $\uparrow$
Overall	156,376	1,397	329	341	80.94	80.38	80.66
Multi	51,636	633	171	197	78.73	76.27	77.48
(b) L2 action prediction							
Occupancy	Gold action	Decision points	EM $\uparrow$	Launch P/R/F1 $\uparrow$	Cancel P/R/F1 $\uparrow$	Invalid $\downarrow$	
Empty	Keep	34,761	92.5	–	–	–	1.72
Empty	Launch	7,062	78.3	82/77/79	–	–	0.30
Single	Keep	40,238	86.7	–	–	–	5.15
Single	Launch	8,917	70.2	78/72/75	–	–	0.84
Single	Cancel	9,846	58.4	–	–	70/61/65	1.02
Single	C+L	3,916	49.3	74/68/71	66/57/61	–	1.65
Multi (2–3)	Keep	24,713	78.1	–	–	–	8.11
Multi (2–3)	Launch	5,842	61.1	71/64/67	–	–	5.56
Multi (2–3)	Cancel	9,276	52.8	–	–	58/46/51	6.71
Multi (2–3)	C+L	5,453	37.4	66/58/62	52/40/45	–	7.23
Full (4)	Keep	3,411	74.4	–	–	–	15.09
Full (4)	Cancel	2,116	48.6	–	–	54/42/47	6.35
Full (4)	C+L	825	33.9	61/53/57	48/35/41	–	13.28
(c) L2 invalid-action breakdown							
Scope	Other occupied launch	Other idle cancel	Duplicate launch	Duplicate cancel	Total		
Overall	1,000	570	410	260	2,240		
Multi	860	460	350	215	1,885		
Full	150	75	65	35	325		

The shared evaluation set contains 1,000 episodes and 156,376 decision points. Precision, recall, F1, EM, and Invalid are percentages; decision points, TP, FP, FN, and panel (c) are counts. C+L denotes cancel+launch. P/R/F1 match the action token and slot; EM executes the sequence in order and permits permutation among initially idle slots. Invalid tokens count as false positives and make EM zero. Panel (b) uses Multi (2–3); Multi in panels (a,c) means at least two in-flight requests and includes Full. In panel (c) each invalid token gets one category, with duplicate errors preceding occupied-launch or idle-cancel errors and Full occupying all four slots.

the episodes into 156,376 jointly labelled L1/L2 decision points and uses gold prefixes. Information utilization instead supplies gold memory or Tool-Block context, so it scores the use of information rather than the decision to request it. Table 17 reports the action decisions and Table 20 the utilization results.

Table 18. Complete FDB-v1 turn-taking and FDB-v1.5 overlap-handling results.

Task	Metric	Reference systems					AdaptDuplex training						
		Moshi [†]	Freeze-Omni [†]	Qwen3-Omni +VAD	MiniCPM -o 4.5	Duplex Omni	Mixed SFT	Curriculum				+ Joint SFT	+ RL (GRPO)
								Fixed 0.48 s	Fixed 0.64 s	Fixed 0.96 s	Dynamic		
(a) FDB-v1: turn-taking													
Pause	Syn. TOR↓	0.985	0.642	0.891	0.270	0.116		0.131	0.130	0.125	0.128	0.126	0.123
	Can. TOR↓	0.980	0.481	0.935	0.463	0.097		0.431	0.406	0.378	0.299	0.291	0.284
Back-channel	TOR↓	1.000	0.636	0.964	0.273	0.491		0.346	0.335	0.327	0.334	0.331	0.327
	Freq.↑	0.001	0.001	0.026	0.020	0.034		0.041	0.040	0.039	0.040	0.040	0.041
	JSD↓	0.957	0.997	0.871	0.902	0.847		0.824	0.823	0.818	0.821	0.818	0.815
Smooth turn	TOR↑	0.941	0.336	0.992	0.882	0.193		0.992	0.980	0.958	0.982	0.984	0.985
	Resp. lat.↓	0.265	0.953	8.454	1.222	0.000 [‡]		0.052	0.079	0.112	0.085	0.079	0.076
Inter-ruption	TOR↑	1.000	0.867	0.995	0.985	0.746		0.985	0.983	0.985	0.985	0.986	0.986
	GPT-4o↑	0.765	3.615	4.899	4.497	4.222		4.168	4.174	4.216	4.512	4.543	4.570
	Resp. lat.↓	0.257	1.409	9.582	1.408	0.158		0.104	0.163	0.234	0.131	0.119	0.113
(b) FDB-v1.5: overlap handling													
Interruption	Respond↑	0.50	0.72	0.97	0.82	0.67	0.62	0.82	0.78	0.76	0.79	0.83	0.85
	Resume↓	0.26	0.12	0.00	0.10	0.12	0.13	0.10	0.11	0.12	0.11	0.09	0.08
	Uncertain↓	0.00	0.03	0.02	0.01	0.00	0.01	0.02	0.00	0.00	0.01	0.00	0.00
	Unknown↓	0.25	0.13	0.01	0.07	0.22	0.24	0.08	0.11	0.12	0.09	0.08	0.07
	Stop lat.↓	1.16	1.42	0.49	2.49	2.01	2.86	1.19	1.81	2.61	1.32	1.20	1.08
	Resp. lat.↓	1.47	1.35	10.28	1.86	0.36	1.25	0.33	0.42	0.58	0.34	0.31	0.27
Backchannel	Respond↓	0.02	0.07	0.30	0.01	0.01	0.00	0.01	0.01	0.01	0.01	0.00	0.00
	Resume↑	0.06	0.80	0.00	0.46	0.28	0.41	0.52	0.54	0.56	0.55	0.55	0.57
	Uncertain↓	0.00	0.02	0.24	0.00	0.02	0.01	0.01	0.00	0.00	0.00	0.00	0.01
	Unknown↓	0.92	0.11	0.46	0.53	0.69	0.58	0.46	0.45	0.43	0.44	0.45	0.42
	Stop lat.↑	0.42	0.66	–	0.76	0.80	0.86	0.82	0.84	0.85	0.84	0.86	0.89
	Resp. lat.↓	3.00	2.16	4.37	2.34	0.82	1.77	0.32	0.66	1.14	0.43	0.32	0.31
Talking to others	Respond↓	0.20	0.58	0.97	0.64	0.15	0.47	0.53	0.51	0.50	0.50	0.47	0.46
	Resume↑	0.19	0.25	0.00	0.20	0.36	0.26	0.29	0.30	0.33	0.33	0.35	0.41
	Uncertain↑	0.02	0.00	0.00	0.02	0.01	0.03	0.04	0.05	0.01	0.03	0.04	0.05
	Unknown↓	0.59	0.15	0.03	0.14	0.47	0.24	0.14	0.14	0.16	0.14	0.14	0.08
	Stop lat.↑	0.87	1.39	0.14	1.71	1.87	1.89	1.88	1.90	1.95	1.89	1.91	1.87
	Resp. lat.↓	2.38	1.32	6.16	1.85	1.95	1.53	0.32	0.32	0.34	0.33	0.32	0.29
Background speech	Respond↓	0.21	0.63	1.00	0.43	0.05	0.47	0.50	0.44	0.37	0.42	0.40	0.36
	Resume↑	0.07	0.25	0.00	0.36	0.07	0.23	0.33	0.42	0.46	0.42	0.45	0.42
	Uncertain↑	0.01	0.01	0.00	0.02	0.00	0.01	0.02	0.01	0.00	0.01	0.02	0.06
	Unknown↓	0.71	0.11	0.00	0.19	0.89	0.29	0.15	0.13	0.17	0.15	0.13	0.16
	Stop lat.↑	0.54	0.98	0.20	1.46	1.50	1.46	1.44	1.47	1.59	1.51	1.56	1.60
	Resp. lat.↓	1.62	1.60	6.56	1.61	0.74	2.07	0.30	0.39	0.59	0.36	0.33	0.31

(a); Freq. in events/s; GPT-4o scores 0–5; latencies in seconds. GPT-4o score and all latencies conditional on TO=1. Mixed SFT is not evaluated on FDB-v1.

[‡]Negative smooth-turn latencies clipped to zero; excluded from best-value highlighting. [†]From FDB-v1 [14] Table III; latencies not hardware-matched.(b): Respond/Resume/Uncertain/Unknown are rates (0–1); Stop/Response are latencies (s). –: no qualifying event. [†]From FDB-v1.5 [15] Table 2; latencies are not hardware-matched.

Table 19. Execution statistics for serialization formats and window policies.

Format	Window policy	Group	Workload		Comp. Cost		Exec. Time (ms)				
			Windows	Input (min)	GPU time (s)	GPU-s/min ↓	Mean	P50↓	P90↓	P95↓	P99↓
JSON-style	Fixed 0.48 s	All	3,724	29.792	2,833.331	95.104	760.8	789.4	957.6	1,018.7	1,164.2
		SPEAK	2,503	20.024	2,177.410	108.740	870.0	844.8	1,041.2	1,116.5	1,269.3
		LISTEN	1,221	9.768	655.921	67.150	537.2	529.6	658.4	711.8	879.4
Canonical	Fixed 0.48 s	All	3,724	29.792	1,439.586	48.321	386.6	418.5	491.2	565.9	672.2
		SPEAK	2,503	20.024	1,127.973	56.331	450.6	433.1	527.9	605.5	689.9
		LISTEN	1,221	9.768	311.613	31.901	255.2	254.3	316.7	359.7	506.7
	Fixed 0.64 s	All	2,790	29.760	1,223.506	41.112	438.5	480.1	581.7	638.5	741.8
		SPEAK	1,901	20.277	985.681	48.610	518.5	500.5	614.6	678.5	768.6
		LISTEN	889	9.483	237.825	25.080	267.5	265.5	338.9	384.0	538.3
	Fixed 0.96 s	All	1,856	29.696	993.492	33.455	535.3	595.3	714.6	773.3	868.8
		SPEAK	1,298	20.768	827.248	39.833	637.3	619.1	745.8	802.3	901.2
		LISTEN	558	8.928	166.244	18.621	297.9	293.7	381.1	440.8	613.1
	Dynamic	All	2,430	29.712	1,304.880	43.920	465.2	492.6	603.8	665.1	781.4
		SPEAK	1,660	20.064	1,056.420	52.615	548.7	541.3	658.9	721.6	836.5
		LISTEN	770	9.648	248.460	25.751	285.7	286.4	354.2	399.8	566.9

Input is the duration attributed to each group. Cost = GPU seconds / input minutes; ALL aggregates SPEAK and LISTEN. Exec. time is distinct from interaction latency.

Table 20. L1 and L2 information utilization.

(a) L1 memory utilization						
Condition	Fact correct/N	Fact Acc.↑	Pref. correct/N	Pref. Acc.↑	FalseUpd. errors/N	FalseUpd. ↓
Rolling only	41/400	10.25	67/400	16.75	–	–
Gold P + M	251/400	62.75	237/400	59.25	17/200	8.50
(b) L2 external-reasoning utilization						
Condition	N	Correct	Behavior Acc.↑	Missing target	Forbidden value	
Single	200	173	86.50	27	0	
Bind	200	147	73.50	31	27	
Compose	200	137	68.50	63	8	
Fail	200	163	81.50	21	20	
None	200	161	80.50	–	39	
(c) Bind counterfactual intervention						
Condition	Pairs	Original Acc.↑	Swapped Acc.↑	Pair Acc.↑		
Bind	200	73.50	70.50	64.00		

All accuracies and FalseUpd. are percentages; other entries are counts. Panel (a) uses 1,000 instances: 400 fact, 400 preference, and 200 no-update cases; the two context conditions are paired on the fact and preference instances. Panel (b) uses 1,000 scenarios, 200 per condition. Behavior Acc. equals one iff all target canaries and no forbidden canaries appear. Missing-target and forbidden-value errors may co-occur. Panel (c) is a paired intervention on the 200 Bind cases: A/B payloads are exchanged while the dialogue, query, and request identities remain fixed, Pair Acc. requires both the original and swapped members to be correct.